

Exercises 1.4

1. (a) $C_5 \times C_{11} \cong C_{55}$, (b) $C_2 \times C_3 \times C_8 \cong C_2 \times C_{24}$,
 (c) $C_3 \times C_8 \times C_4 \times C_9 \cong C_{12} \times C_{72}$,
 (d) $C_5 \times C_6 \times C_7 \times C_8 \cong C_5 \times C_2 \times C_3 \times C_7 \times C_8 \cong C_2 \times C_{840}$.
2. (a) $10 = 2 \times 5$. The only abelian group is $C_2 \times C_5$, which is isomorphic to C_{10} .
 (b) $18 = 2 \times 3^2$. The abelian groups are $C_2 \times C_3 \times C_3 \cong C_3 \times C_6$
 and $C_2 \times C_9 \cong C_{18}$.
 (c) $45 = 3^2 \times 5$. The abelian groups are $C_5 \times C_3 \times C_3 \cong C_3 \times C_{15}$
 and $C_5 \times C_9 \cong C_{45}$.
 (d) $120 = 2^3 \times 3 \times 5$. The abelian groups are $C_2 \times C_2 \times C_2 \times C_3 \times C_5 \cong C_2 \times C_2 \times C_{30}$,
 $C_2 \times C_4 \times C_3 \times C_5 \cong C_2 \times C_{60}$ and $C_8 \times C_3 \times C_5 \cong C_{120}$.
 (e) $540 = 2^2 \times 3^3 \times 5$. The abelian groups are $C_2 \times C_2 \times C_3 \times C_3 \times C_3 \times C_5 \cong C_3 \times C_6 \times C_{30}$,
 $C_2 \times C_2 \times C_3 \times C_9 \times C_5 \cong C_6 \times C_{90}$, $C_2 \times C_2 \times C_{27} \times C_5 \cong C_2 \times C_{270}$,
 $C_4 \times C_3 \times C_3 \times C_3 \times C_5 \cong C_3 \times C_3 \times C_{60}$, $C_4 \times C_3 \times C_9 \times C_5 \cong C_3 \times C_{180}$,
 $C_4 \times C_{27} \times C_5 \cong C_{540}$.
3. There is only one group of order 11 and one of order 13 (C_{11} and C_{13}).
 The abelian groups of order 12 are C_{12} and $C_2 \times C_6$.
 A_4 , D_6 and Q_6 are non-abelian, of order 12. There are five groups of order 12.
4. Abelian: $C_2 \times C_2 \times C_6$, $C_2 \times C_{12}$, C_{24} (or equivalent). Non-abelian: we know D_{12} , Q_{12}
 and S_4 have order 24. We also have direct products of groups of order 2, 3, 4 with non-
 abelian groups of order 12, 8, 6 respectively: $C_2 \times A_4$, $C_2 \times D_6$, $C_2 \times Q_6$, $C_3 \times D_4$, $C_3 \times$
 Q_4 , $C_4 \times D_3$, $V \times D_3$ (but it turns out that $V \times D_3 \cong C_2 \times D_6$). There are three more
 which can be found in lists of groups.
5. (a) One group: $C_{p_1 p_2 \dots p_r}$ (since all the primes are coprime).
 (b) $k = 1$: one group, C_p . $k = 2$: two groups, $C_p \times C_p$ and C_{p^2} .
 $k = 3$: three groups, $C_p \times C_p \times C_p$, $C_p \times C_{p^2}$ and C_{p^3} .
 Introduce a notation: let $(a, b, c, \dots)$ stand for $C_{p^a} \times C_{p^b} \times C_{p^c} \times \dots$, so the above
 result for $k = 3$ is written as $(1, 1, 1)$, $(1, 2)$ and (3) .
 $k = 4$: five groups $(1, 1, 1, 1)$, $(1, 1, 1, 2)$, $(1, 3)$, $(2, 2)$, (4) .
 $k = 5$: seven groups $(1, 1, 1, 1, 1)$, $(1, 1, 1, 1, 2)$, $(1, 1, 1, 3)$, $(1, 4)$, $(1, 2, 2)$, $(2, 3)$, (5) .
 $k = 6$: eleven groups $(1, 1, 1, 1, 1, 1)$, $(1, 1, 1, 1, 1, 2)$, $(1, 1, 1, 1, 3)$, $(1, 1, 1, 4)$, $(1, 5)$,
 $(1, 1, 2, 2)$, $(1, 2, 3)$, $(2, 2, 2)$, $(2, 4)$, $(3, 3)$, (6) .
 You might now make a conjecture about the number when $k = 7$. Don't be too sure!
 List the groups, and see what you can discover about this sequence (the number of
 unordered partitions of k).
6. (a) The required order is $\text{lcm}(q_1, \dots, q_s)$.
 (b) We know G is isomorphic to a direct product $C_{q_1} \times \dots \times C_{q_s}$. By (a), G has a
 non-identity element of order $\text{lcm}(q_1, \dots, q_s)$. This order must equal 2, so $q_i = 2$ for
 each i . Then $|G| = |C_2|^s = 2^s$.